



Flutter

NUS Orbital: Mobile App Development (Flutter)

Building High-Performance Cross-Platform Apps

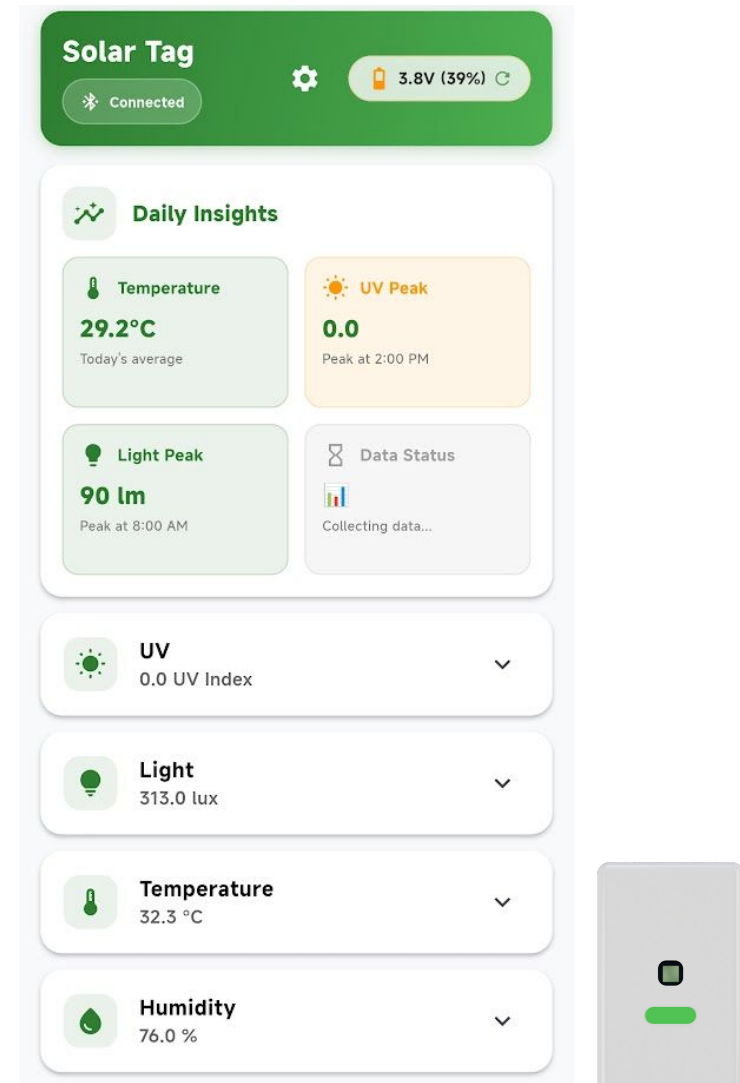
About Me

Kwa Jian Quan

CEG Year 3 (Poly)

 Flutter & Cloud Dev at NOC Internship

 NUS VIP Companion App (UV Tracker)



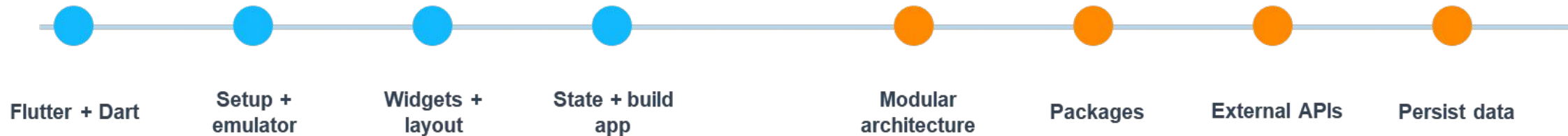
Workshop Schedule

Part 1

Core Flutter concepts

Part 2

Real-world app integration



| Why Flutter?



Unified Code

Write once and deploy
natively to iOS, Android,
Web, and Desktop.



Hot Reload

See code changes while
keeping app state during
development.



Flexible UI

Rich set of Material and
Cupertino widgets.

| Setting Expectations

-  **Concept First:** We focus on the "Flutter way" of thinking, not just memorising syntax or build configurations.
-  **Beginner Friendly:** No prior mobile experience required. However, due to time constraints, it will be relatively fast paced.
-  **Outcome:** By the end, you'll have a clear roadmap to take an idea from wireframe to a functional cross-platform app.

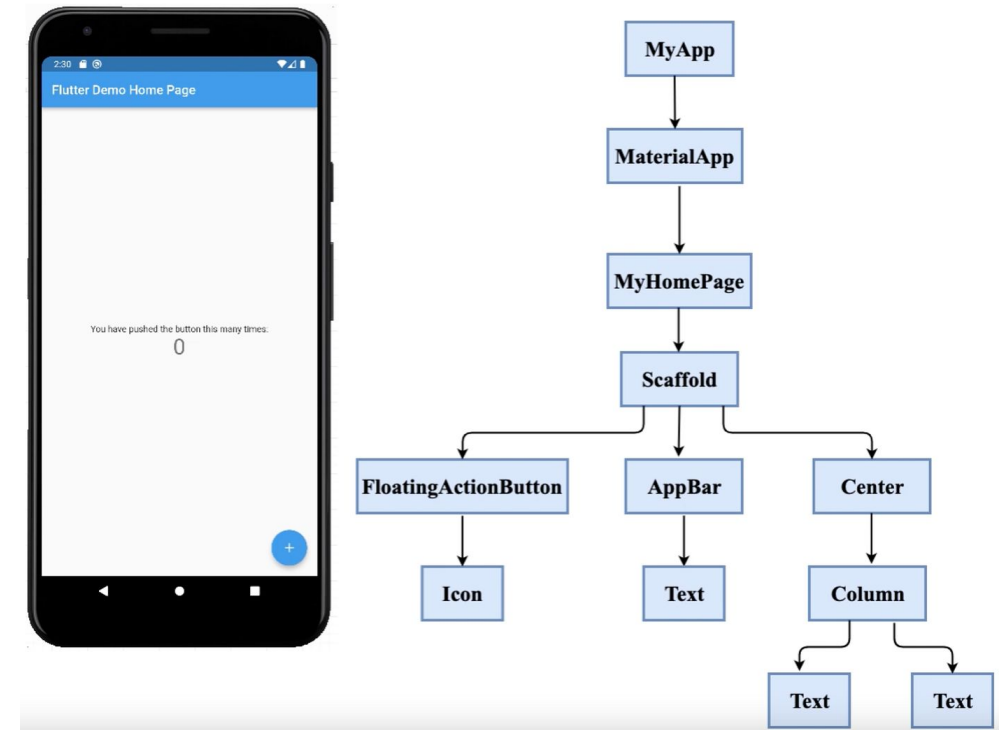
| Widgets

Everything is a Widget

In Flutter, the entire UI is a hierarchy of objects. From simple text to complex layout grids.

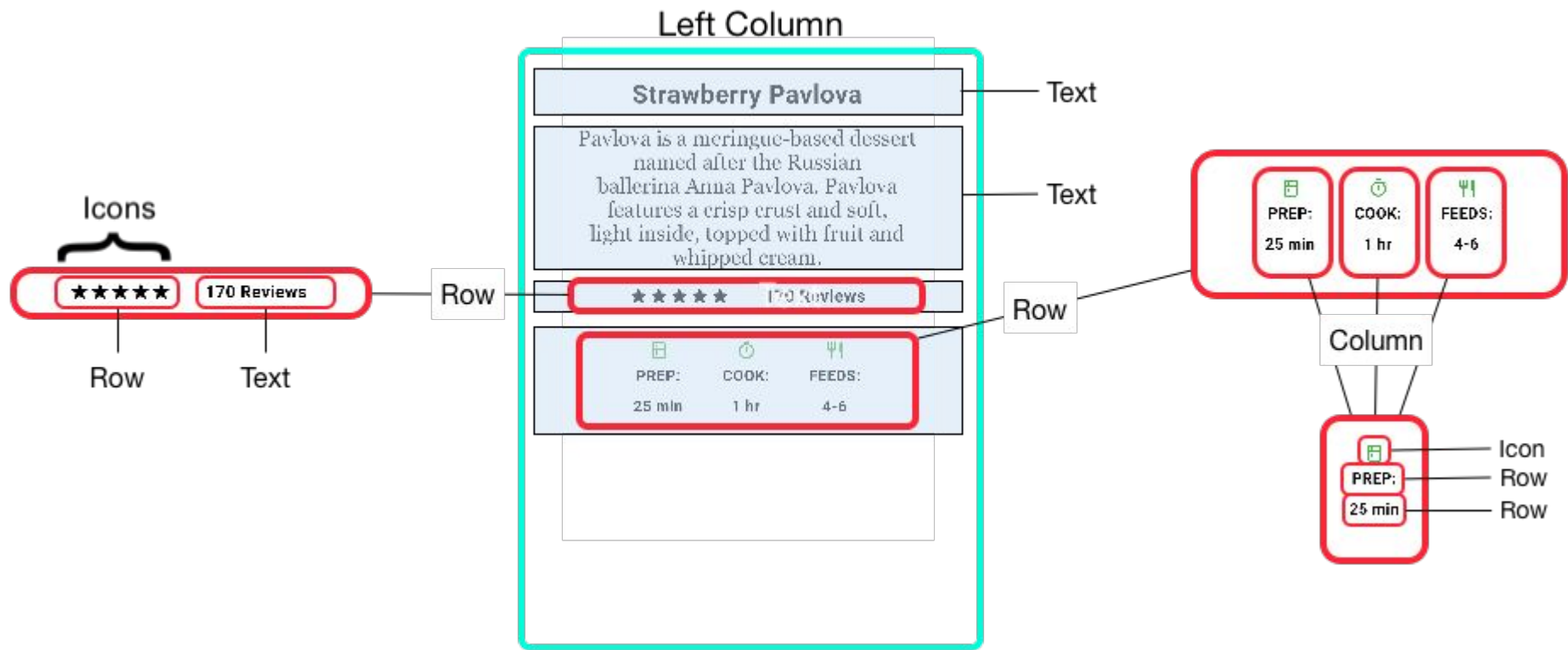
The Tree Structure:

Widgets are nested inside parent widgets. During state changes, Flutter recalculates the tree to update the screen.



<https://medium.com/@ffarzad45/understanding-the-flutter-widget-tree-20d3ed648dbe>

Layout Fundamentals



<https://docs.flutter.dev/ui/layout>

| State Management

2

Core Widget Types

Stateless vs. Stateful

StatelessWidget: The content is immutable. It doesn't change after the widget is built.

StatefulWidget: Can maintain internal data that changes over time. Uses `setState()` to trigger UI rebuilds when data updates.